SLIIT
Discover Your Future

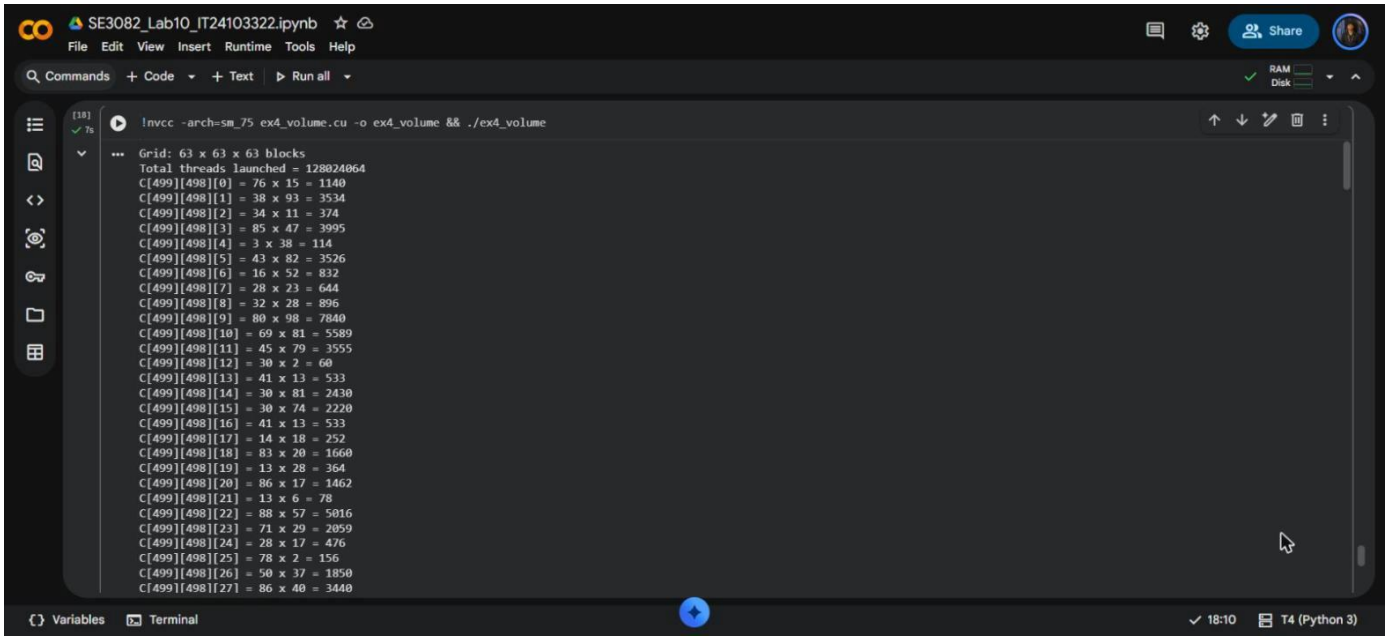
BSc (Hons) in Computer Science
Year 3

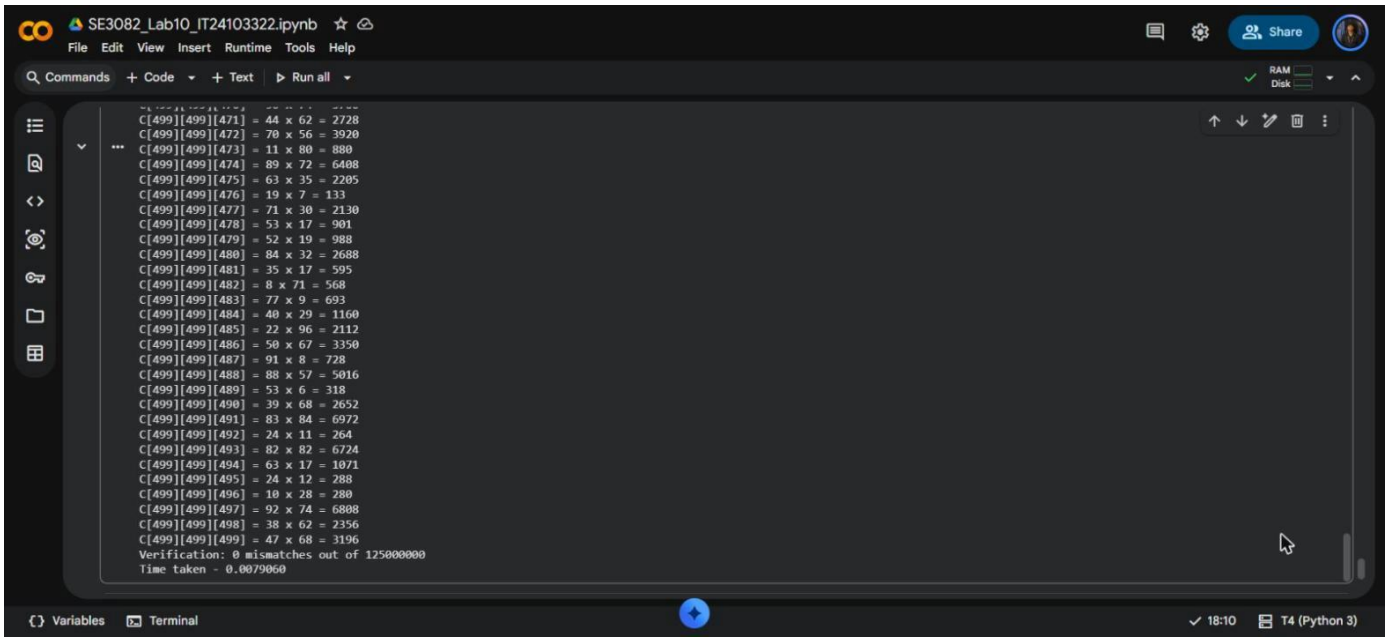
Lab Sheets

SE3082 – Parallel Computing **Semster 1, 2026**

IT24103322

Exercise 4





The volume is 500 × 500 × 500 = 125,000,000 ints (500 MB per matrix), stored as flat arrays. Each block is dim3(8, 8, 8) = 512 threads, and the grid is rounded up in all three directions: 500 / 8 = 62.5 → 63. In the kernel x, y and z each come from blockIdx * blockDim + threadIdx, the element is computed only if x < nx, y < ny and z < nz, and the flat index is (z * ny + y) * nx + x. The last 1000 elements were printed in memory order.

Item	Expected value	My result
Grid	63 x 63 x 63 blocks	63 x 63 x 63 blocks
Total threads launched	128,024,064	128024064
First of the last 1000	C[499][498][0] = 76 x 15 = 1140	C[499][498][0] = 76 x 15 = 1140
Last result	C[499][499][499] = 47 x 68 = 3196	C[499][499][499] = 47 x 68 = 3196
Verification	0 mismatches out of 125000000	0 mismatches out of 125000000

1. **Compare the time with Lab 9 Exercise 6. Is the 3D kernel faster, slower or about the same?**

   The 3D kernel took 0.0079060 s for 125 million elements; the 2D kernel took about 0.00504 s for 100 million. Per element the 3D kernel is about 25 % slower (63 ps against 50 ps). The work per element is the same, so the difference most likely comes from the thread layout: with 8 threads in x, one warp of 32 threads covers 4 short rows of 8 numbers instead of 32 numbers next to each other, so its memory accesses are less coalesced, and each thread also does more index arithmetic.

The notebook SE3082_Lab10_IT24103322.ipynb in my Google Drive contains the working cells for every exercise with the outputs visible.